

RESULTS

*Major Research Project
Master of Science in Data Science and Analytics*

Machine Learning-Based Auto Insurance Fraud Detection

Name: Meshwa Patel
Student Number: 501390663
Supervisor: Shengkun Xie
Toronto Metropolitan University

1 Introduction

This chapter presents and discusses the results of the modelling and testing stages of the project. It begins with the comparison of the four classifiers under both imbalanced and resampled training, identifies and evaluates the best model on the held-out test set, and then reports the explainability analysis, the rule-based fraud risk score, the fairness assessment, and the external validation. The chapter closes with a set of advanced analyses that examine the results more rigorously, and a discussion of what the findings mean. All numbers and figures reported here are taken directly from the executed project notebooks, so the chapter and the code are fully consistent.

1.2 Model Comparison Results

Four classifiers were each trained with and without the Synthetic Minority Over-sampling Technique (SMOTE), producing eight configurations. Table 1.1 reports their performance on the validation set. Because the data is highly imbalanced, the models were judged on precision, recall, F1-score, and AUC-ROC rather than accuracy.

Model	Data	Precision	Recall	F1	AUC
Logistic Regression	Original	0.364	0.016	0.030	0.698
Logistic Regression	SMOTE	0.207	0.612	0.310	0.695
Random Forest	Original	0.333	0.004	0.008	0.740
Random Forest	SMOTE	0.434	0.045	0.081	0.736
XGBoost	Original	0.440	0.099	0.161	0.731
XGBoost	SMOTE	0.431	0.103	0.166	0.731
Neural Network	Original	0.000	0.000	0.000	0.603
Neural Network	SMOTE	0.218	0.357	0.271	0.654

Table 1.1: Validation performance of all eight model configurations

Two clear patterns emerge. First, without SMOTE every model has near-zero recall, effectively predicting that almost no claim is fraudulent; the imbalanced Logistic Regression catches only 1.6% of fraud. With SMOTE, recall rises sharply, most dramatically for Logistic Regression (from 0.016 to 0.612). At the same time, AUC-ROC barely changes when SMOTE is applied, which shows that SMOTE shifts the decision threshold rather than improving the model's underlying ability to rank risk. This agrees with the finding of Ekici (2025) that resampling does not always improve performance.

The model selected by validation F1 was Logistic Regression with SMOTE (F1 = 0.310), which is also the simplest and most interpretable of the four, a useful property for a project centered on explainability. The tree-based models achieved slightly higher AUC but caught far less fraud at the default threshold.

A notable result is that the Neural Network, the most powerful model tested, performed worst (AUC 0.654), and without SMOTE flagged no fraud at all. The fact that the most expressive model could not beat a simple linear one is strong evidence that the ceiling on performance is set by the data rather than by the choice of model.

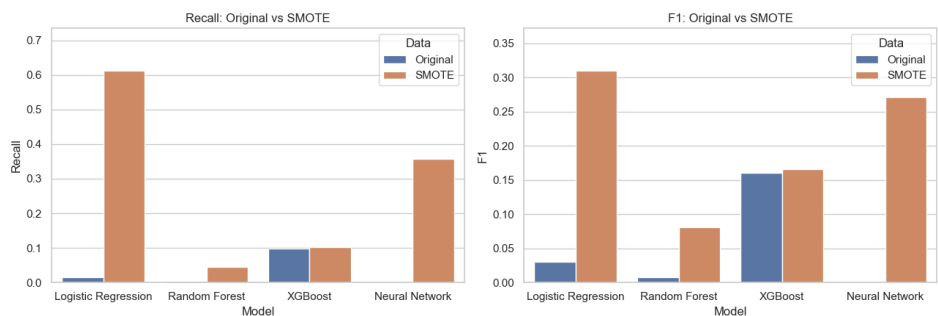


Figure 1.1: The effect of SMOTE on recall and F1 across the four models. SMOTE sharply raises recall, especially for Logistic Regression.

1.3 Best Model on the Test Set

The selected model was evaluated once on the held-out test set of 4,500 claims, which was never used during training or model selection. Table 1.2 reports the result.

Metric	Value	Interpretation
Precision	0.222	About 2 in every 10 flagged claims were genuinely fraudulent.
Recall	0.686	The model caught roughly 7 in every 10 real frauds.
F1-score	0.335	The balance between precision and recall.
AUC-ROC	0.726	Clearly better than random guessing (0.5), but not high.

Table 1.2: Test-set performance of the best model (Logistic Regression + SMOTE).

The test F1 of 0.335 is very close to the validation F1 of 0.310, which indicates the model did not overfit. In practical terms, it flagged 1,596 of the 4,500 claims and caught 354 of the 516 real frauds. The high recall therefore comes at the cost of many false alarms, which is why the model is best understood as a triage tool that ranks claims for investigators rather than an automatic accept-or-reject system.

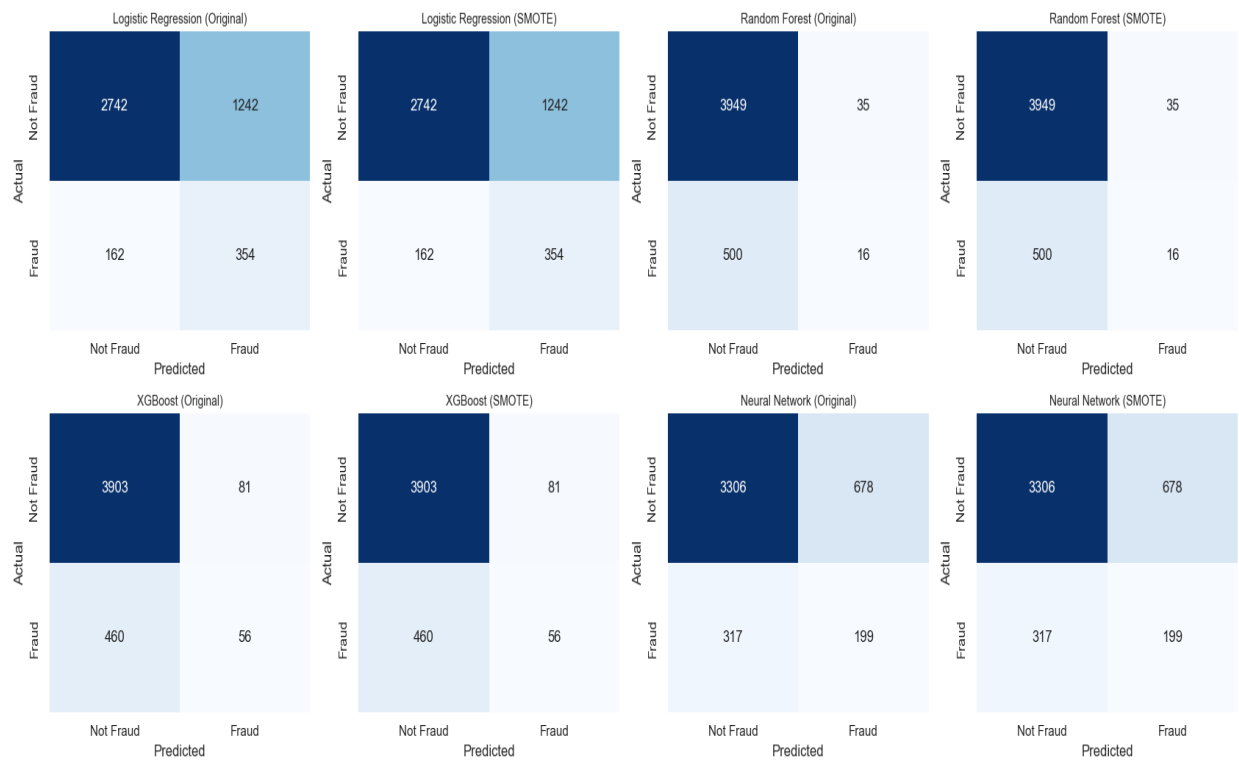


Figure 1.2: Confusion matrices for all eight model configurations on the test set.

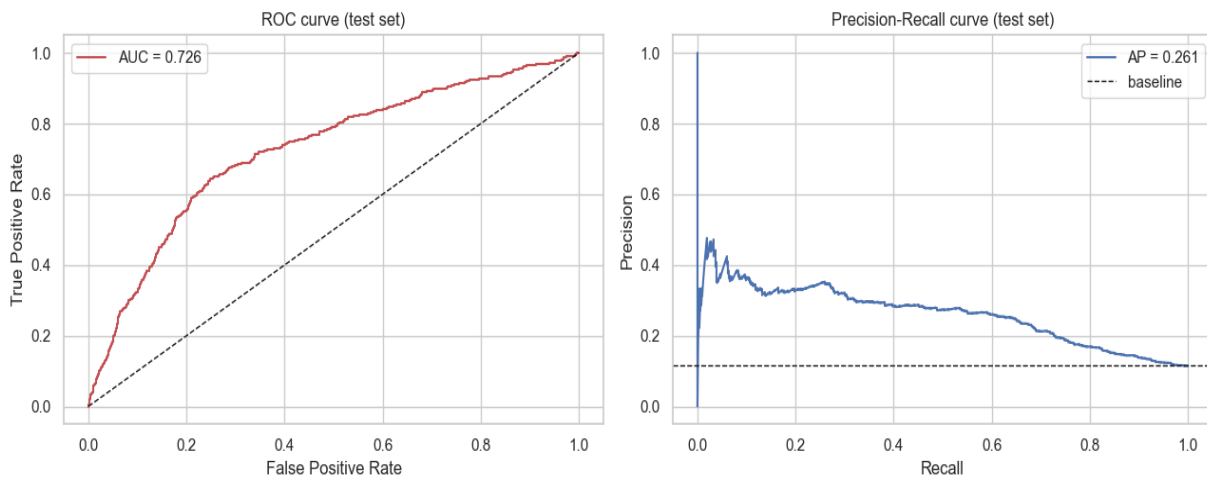


Figure 1.3: ROC curve and Precision-Recall curve for the best model (Logistic Regression + SMOTE) on the test set.

1.4 Cross-Validation and Stability

To confirm that these results were not an artefact of one particular data split, each configuration was re-evaluated using Repeated Stratified K-Fold cross-validation (5 folds, 2 repeats, giving 10 scores per model). Table 1.3 reports the accuracy, F1, and AUC as mean, standard deviation, and 95% confidence interval.

Model (SMOTE)	Accuracy %	F1 %	AUC %
Logistic Regression	69.3 $\pm$ 0.5	32.6 $\pm$ 0.6	71.2 $\pm$ 0.6
Neural Network	79.7 $\pm$ 0.9	30.3 $\pm$ 0.7	67.3 $\pm$ 0.6
XGBoost	87.9 $\pm$ 0.2	18.0 $\pm$ 1.2	75.1 $\pm$ 0.9
Random Forest	88.2 $\pm$ 0.2	7.9 $\pm$ 1.2	75.1 $\pm$ 0.7

Table 1.3: Cross-validated results (mean $\pm$ standard deviation) for the SMOTE models.

Three points stand out. The standard deviations are small (mostly well under 1%), so the results are stable and repeatable rather than lucky. Logistic Regression with SMOTE has the highest F1 (32.6%), and its confidence interval [32.1%, 33.0%] does not overlap with that of the next-best model, so its lead is statistically meaningful. Finally, and importantly, every model trained without SMOTE reaches about 88% accuracy, which is below the 88.5% that a model would score by simply predicting “no fraud” for every claim. This confirms that accuracy is the wrong measure for this problem and justifies the decision to select on F1 instead.

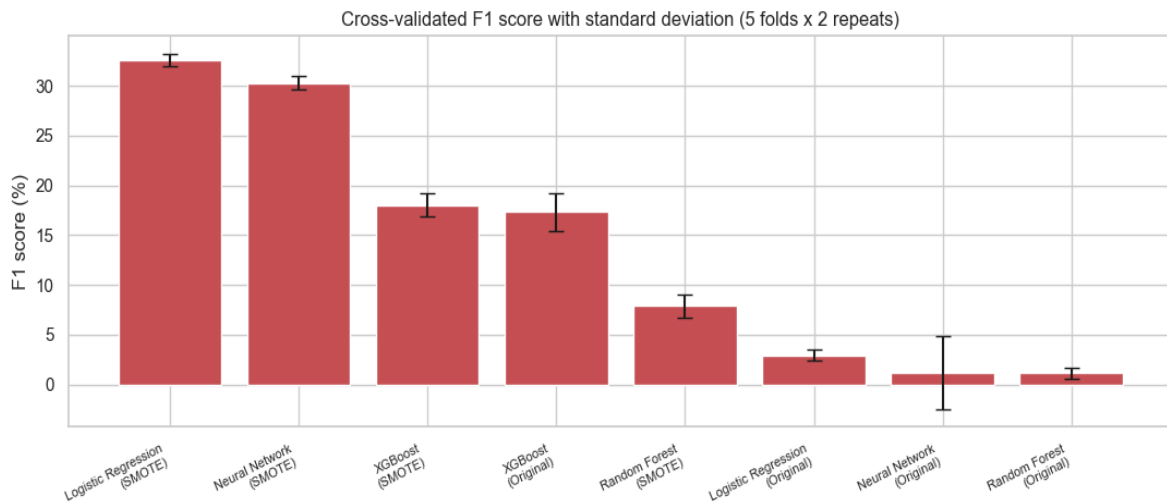


Figure 1.4: Cross-validated F1 with standard-deviation error bars. Short bars indicate stable, repeatable results.

1.5 Explainability and Fraud Risk Scoring

SHAP was applied to the best model to explain its predictions, both overall and for individual claims. The most influential features were the absence of contacted authority, the absence of witnesses, the total-to-claim ratio, and the claimant's age, with no single feature dominating.

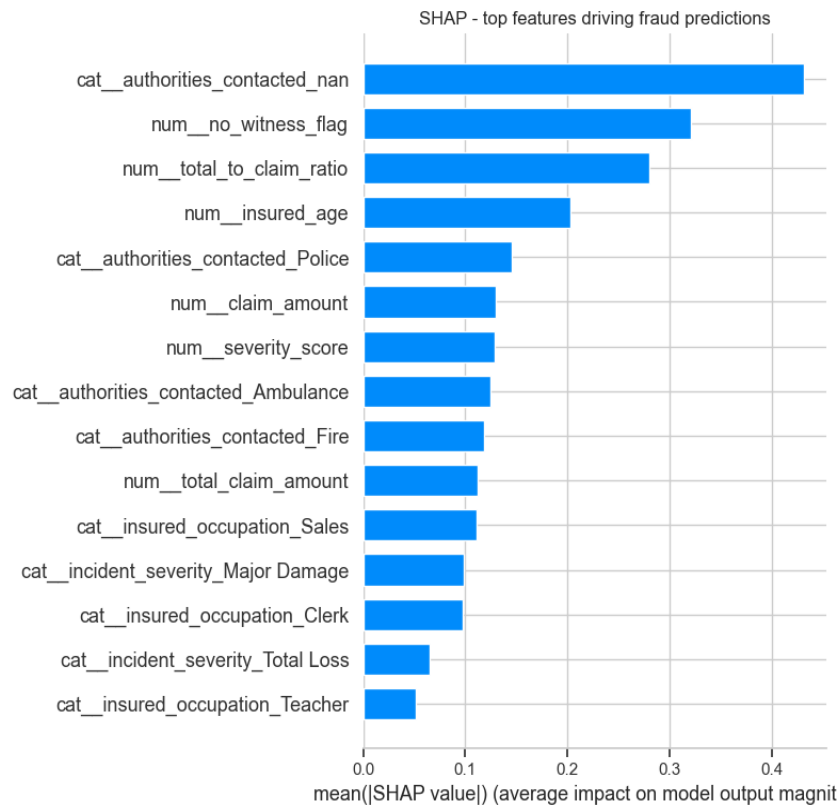


Figure 1.5: SHAP global feature importance for the best model.

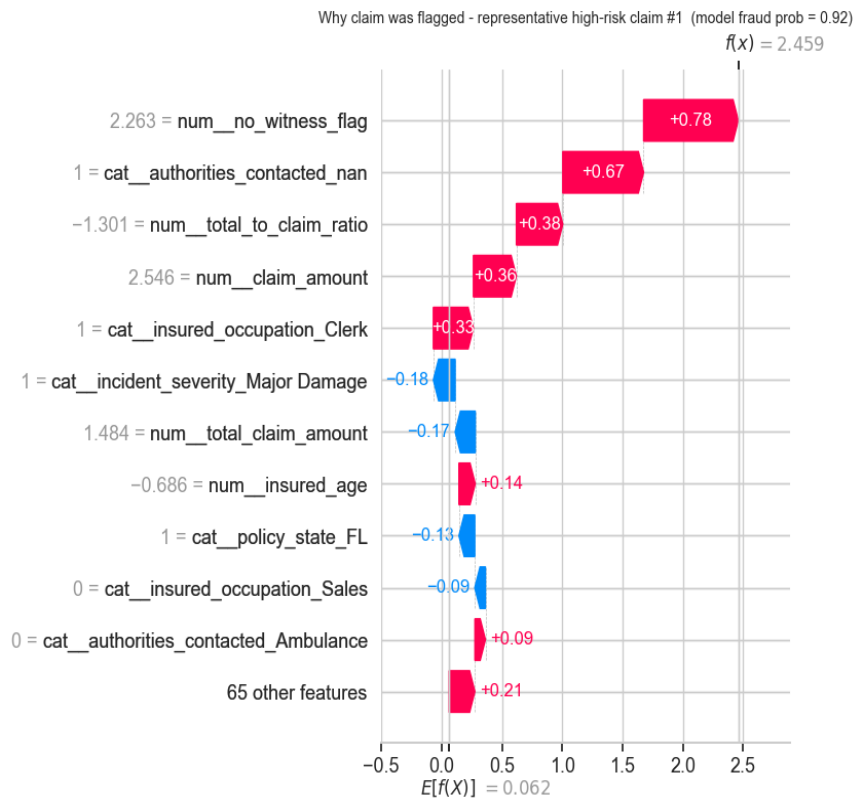


Figure 1.6: SHAP waterfall plot explaining why one individual high-risk claim was flagged.

Because a SHAP chart is still a technical output, a transparent rule-based Fraud Risk Score was also built so that investigators can prioritize claims without running the model. The point weights were not chosen by intuition but derived from the data: each red flag was measured on the training data by its fraud lift, weak flags were dropped, and the surviving flags were given points proportional to their lift. Two flags originally included (“high claim amount” and “no police report”) were removed because their lift showed no real link to fraud. Table 5.4 shows the result on the test set.

Risk tier	Number of claims	Actual fraud rate
Low	2,719	8.79%
Medium	1,256	10.91%
High	525	26.67%

Table 1.4: Actual fraud rate by data-driven risk tier (test set).

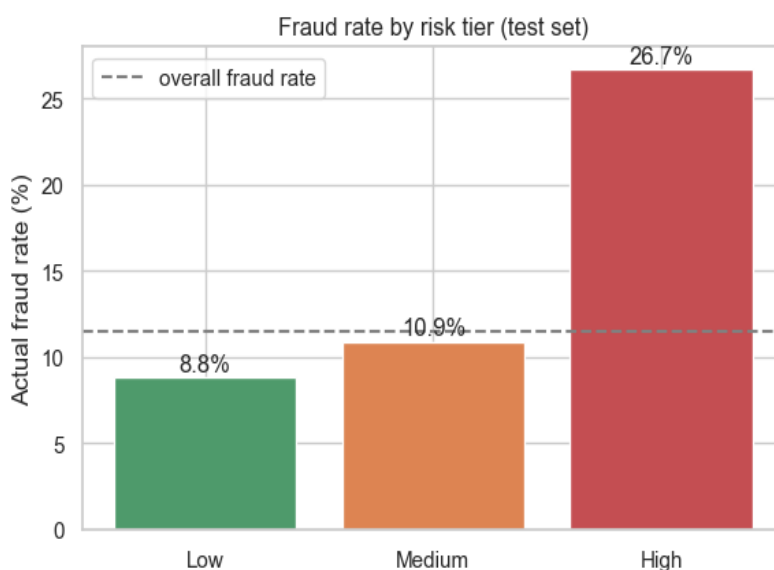


Figure 1.7: Actual fraud rate by risk tier. The fraud rate rises steadily from the Low to the High tier.

The score behaves exactly as intended: the fraud rate rises steadily across the tiers, so the High tier has roughly three times the fraud rate of the Low tier. Correcting the weights from the data (rather than by intuition) markedly improved the score's separation and raised its agreement with the machine learning model's own probabilities. This gives investigators a defensible way to prioritize claims even without the model.

1.6 Fairness Assessment

The model's predictions on the test set were broken down by age, sex, and education. This revealed a bias worth reporting: the model over-flags younger claimants relative to their actual fraud rate, flagging the under-30 group far more often than their true offending rate would justify, while flagging older claimants comparatively less.

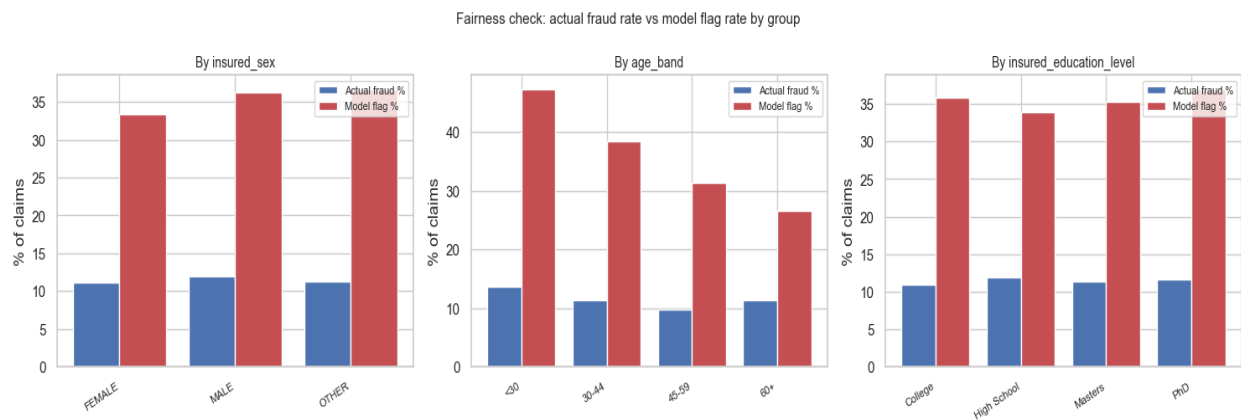


Figure 1.8: Actual fraud rate versus model flag rate by demographic group. The model over-flags younger claimants.

This bias was not only diagnosed but treated. Retraining the model without the demographic features (age, sex, education) brought the flag rates across age groups much closer together and did so at essentially no cost to performance. In fact, the F1-score improved very slightly. This is an important and encouraging result: the model was using age as a shortcut, but it did not need to, because the remaining features carry the same information more fairly. For any real deployment, the demographic-free version would therefore be recommended.

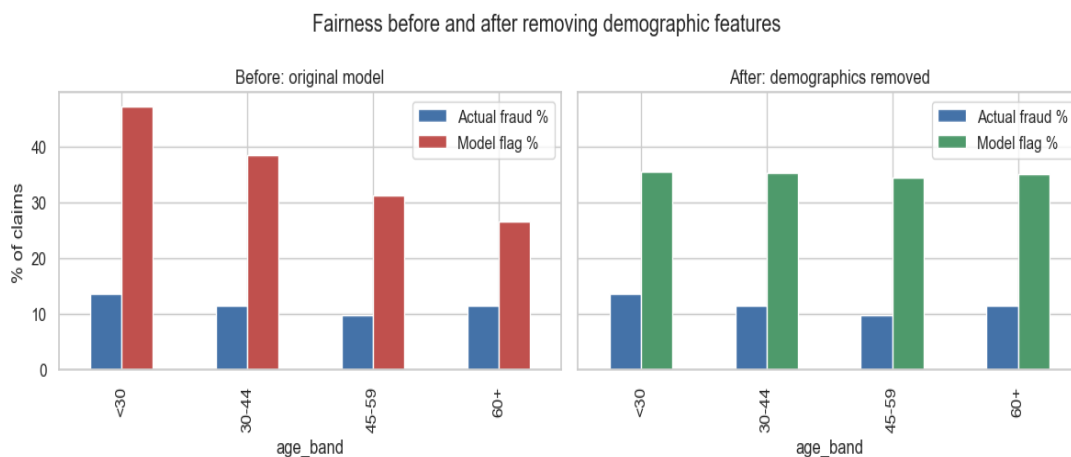


Figure 1.9: Flag rates by age group before and after removing demographic features. The gaps become much more even after the fix.

1.7 External Validation

To test whether the method generalizes, the pipeline was applied to three additional labelled insurance datasets in two ways: a transfer test, where the model trained on the main dataset was applied directly to another dataset, and a benchmark test, where the whole method was retrained from scratch on each dataset. Table 1.5 summarizes the results.

Dataset	Test type	Best AUC	Interpretation
Main dataset	Held-out test	0.726	Honest baseline result.
insurance_claims	Transfer test	0.374	Below chance – does not transfer.
insurance_claims	In-domain check	0.784	The dataset itself has real signal.
fraud_oracle	Benchmark (4 models)	0.855	Method works well on real-signal data.
insurance_fraud_data	Benchmark (4 models)	0.630	Weaker signal but still works.

Table 1.5: Summary of external validation results (key evidence highlighted).

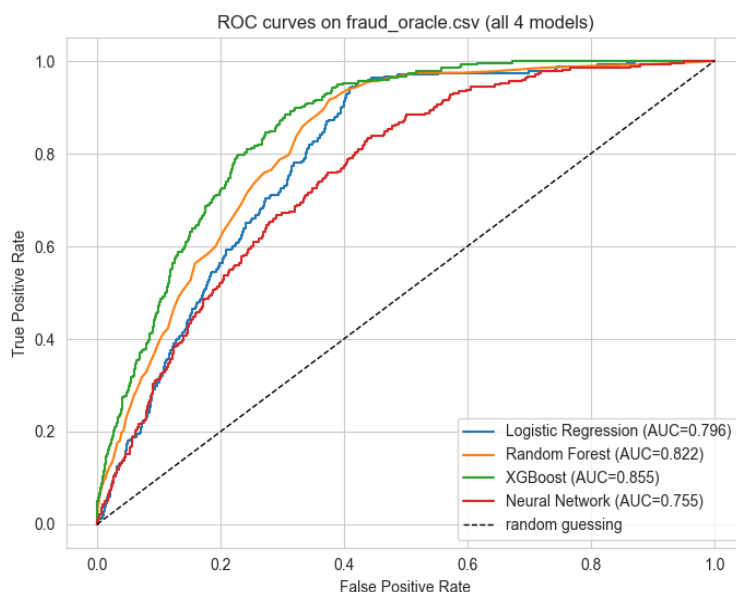


Figure 1.10: ROC curves for the independent benchmark on the fraud_oracle dataset, where the method reached an AUC of 0.855.

The transfer test scored below chance (0.374), yet the same dataset produced a healthy 0.784 when the model was retrained on it. This shows the failure is a domain gap, not a broken method: the main dataset behaves differently from real claims. Most importantly, when the full method was retrained on the fraud_oracle dataset it reached an AUC of 0.855, far higher than on the main data. This is the key cross-dataset evidence that the pipeline is sound and performs well wherever the data carries genuine fraud signal.

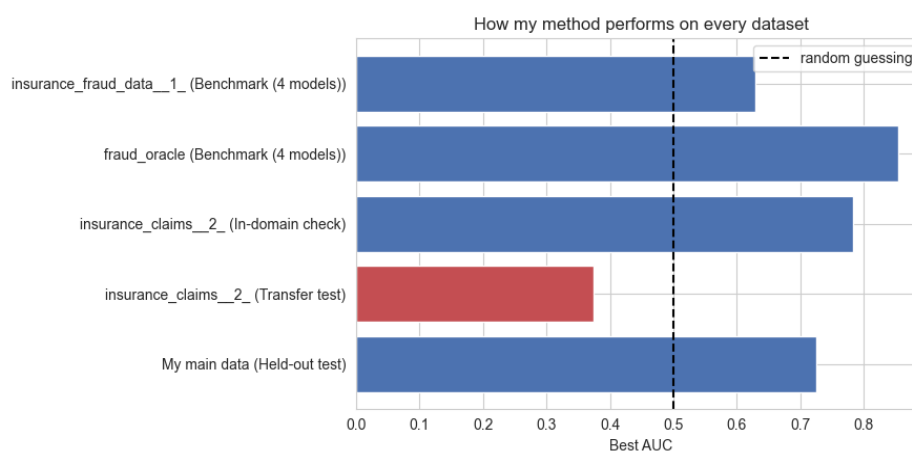


Figure 1.11: Summary of AUC across all external datasets and test types.

1.8 Advanced Analysis

Six further analyses were carried out to examine the results more rigorously than a single score allows.

- **Learning curve.** Test AUC rose from 0.714 at a quarter of the data to only 0.726 at the full amount, then flattened. More data would therefore not help; the limit is the signal in the data, not its quantity.

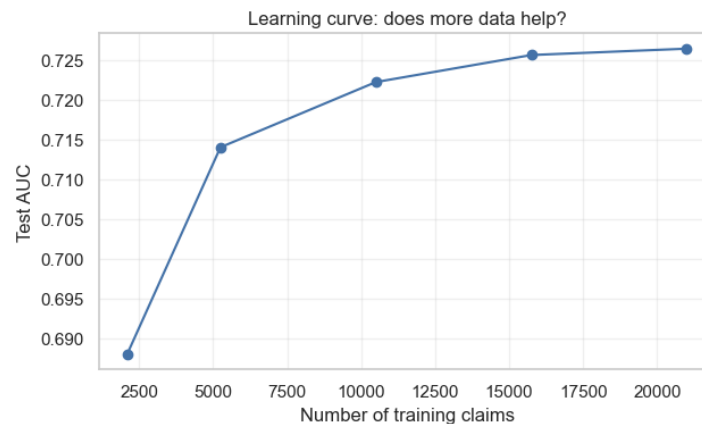


Figure 1.12: Learning curve. Test AUC plateaus well before the full training set is used, showing more data would not help.

- **Ranking value (Precision@K).** Among the 50 claims the model ranked riskiest, 40% were fraud – about 3.5 times better than random. Reviewing the top 1,000 claims would catch 53.5% of all fraud. As a ranking tool, the model is genuinely useful despite its modest AUC.
- **Feature ablation.** Removing the authority information cost the most accuracy (-0.070 AUC), followed by witness information (-0.030). Removing the demographic features costs almost nothing (-0.004), confirming that the fairness fix is essentially free.
- **Statistical significance.** XGBoost AUC advantage over Logistic Regression was real (95% CI [0.019, 0.058], McNemar $p < 0.001$), but small. Statistically significant is not the same as practically important, so Logistic Regression was still preferred for its higher recall and interpretability.
- **Calibration.** The Brier score (0.210) showed the probabilities are inflated by SMOTE, so the model should be used to rank claims rather than to quote a literal probability. The ranking itself remains trustworthy.
- **Business value.** Using an average claim value of about \$12,900, the model produced a net benefit of roughly \$4 million on the 4,500-claim test set. Because a caught fraud is worth far more than a wasted investigation, the low precision matters much less in monetary terms.

1.9 Discussion

Taking together, the results tell a consistent story. On the main dataset the headline numbers are modest: a test AUC-ROC of 0.726, high recall, and low precision. Read correctly, this is an honest result rather than a failure. The exploratory analysis showed a weak, diffuse fraud signal; the most powerful model (the Neural Network) performed worst; the learning curve flattened; and the transfer test fell below chance. Every one of these points to the same conclusion, namely that the main dataset carries very little genuine fraud signal and is most likely synthetic.

The external benchmarks confirm that this is a property of the data and not of the method. The identical pipeline reached an AUC of 0.855 on the fraud_oracle dataset, in line with published studies. The method is therefore sound; it simply needs data with real signal to perform well.

Finally, the project shows that a modest statistical model can still be valuable in practice. As a ranking and triage tool the model makes investigators several times more efficient at the top of the list, its decisions can be explained through SHAP and the transparent risk score, its bias can be removed at no cost, and in monetary terms it would pay for itself many times over. The fairness finding is a reminder that even a modest model requires monitoring before deployment.

1.10 Summary of Findings

- **Best model:** Logistic Regression with SMOTE, test AUC 0.726, F1 0.335. It functions as a claim-ranking tool, not an automatic decision-maker.
- **The Neural Network performed worst**, showing the limit is the data, not the model.
- **Accuracy is unsuitable** for this problem, since no model beats the 88.5% do-nothing baseline on accuracy.
- **The data-driven risk score works**, with fraud rising from 8.8% in the Low tier to 26.7% in the High tier.
- **Bias was found and fixed** by removing demographic features, at no cost to performance.
- **The method is sound:** the same pipeline reached AUC 0.855 on a real-signal dataset, confirming the modest main-dataset results stem from a likely synthetic dataset.